DEPARTMENT OF COMPUTER ENGINEERING
CSD415 MAJOR PROJECT

Academic Year: 2026-27 **Class: S7 C**

Project Approval Form

Date:

Roll No	KTU Reg No	Name	Signature
13	CHN23CS030	ALEESHA MA	
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Submit three project topic proposals with a very small description. At the end of each topic mention two major references of the topic you referred. Refer international journals of repute like IEEE / Elsevier / Springer/ other SCI indexed journals for selecting seminar topic.
(Give the topics in your preference order)

Name of Topic 1: AI-Based Landslide Risk Prediction and Community Early Warning System

Abstract: Kerala faces frequent landslides during monsoon seasons, causing loss of life, property damage, and transportation disruptions. The main stakeholders of this system are residents in landslide-prone areas, disaster management authorities, district administrations, emergency response teams, local self-government bodies, and tourists visiting hilly regions. The proposed AI-based system predicts landslide risks by analyzing rainfall patterns, weather data, historical landslide records, terrain elevation, and slope information. It provides location-based risk alerts through an interactive map and sends early warnings with safety recommendations to help communities and authorities take timely action. The system improves disaster preparedness, supports efficient emergency response, and reduces the impact of landslides

Ref1: https://doi.org/10.1016/j.rines.2025.100146
Ref2: https://ieeexplore.ieee.org/document/10596300

Name of Topic 2: Trust Charge

Abstract: Reliability-Aware EV Charging Navigation with Prepaid Booking & Priority Membership — EV charging apps show “available” stations using data that’s often stale or wrong, and offer no way to reserve a slot or get priority on scarce fast chargers, which fuels range anxiety and slows EV adoption, especially on long routes with few backup options; this project addresses that gap for EV drivers, fleet/business operators, charging network operators, and transport authorities through a system combining crowd-sourced, time-decayed reliability scoring, arrival-time availability prediction, prepaid slot booking, and an Alpha Membership tier for priority fast-charging access (with a fair-use rule so regular users aren’t blocked when there’s no contention), ultimately making charging trustworthy and reservable, giving operators predictable demand and new revenue, and supporting Kerala’s growing EV infrastructure.

Ref1: https://ieeexplore.ieee.org/document/10625267
Ref2: https://ieeexplore.ieee.org/document/9346989

Name of Topic 3: CommuteCram – AI-Based Micro-Learning Podcast Generator

Abstract:AI-Based Micro-Learning Podcast GeneratorMany students spend their daily commute scrolling through social media instead of revising their studies, leading to inefficient use of time and poor retention of lecture content. This problem affects college students who travel by bus, train, or on foot every day. Since commuting is a regular activity, converting this idle time into productive learning can significantly improve academic performance. A software solution can allow students to upload lecture notes or PDFs, use AI to generate concise summaries, convert them into personalized audio podcasts using text-to-speech, and adjust the podcast length based on commute duration while asking voice-based quiz questions at the end. This improves learning efficiency, increases knowledge retention, and encourages consistent revision through gamification.

Ref1: https://doi.org/10.1109/ACCESS.2019.2899655
Ref2:https://doi.org/10.1109/ACCESS.2020.2988510

Guide's Name :

Guide's Suggestion (if any**):**

Programme Assessment Committee (PAC):

1. Approved/Rejected 2. Approved/Rejected 3. Approved/Rejected
Selected Topic:

Project Co-Ordinator HOD